

LESSON 10.8

EXPLORING FUNCTIONS USING EQUATIONS, TABLES, AND GRAPHS



LESSON INTRODUCTION

MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

LEARNING TARGETS

- I can create a table of values from the equation of a function.
- I can graph a function using a table of values.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input/output table, determine a function type that could represent it.

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real-number system.

ESSENTIAL QUESTIONS

- How do you complete a table of values from an equation?
- How do you graph a function using a table of values?
- How do you know if an equation is linear or non-linear?

LESSON NARRATIVE

This lesson begins by revisiting squares and cubes and creating tables of values used to match equations to their graphs. Students will explore and interact with different equations, tables, and graphs (linear, quadratic, and exponential). After making observations, students will write a conjecture of what the equation of a linear function looks like.

COMPONENT SUMMARY

10.8.1 WARM-UP (~5 MIN.)

Support arguments with an explanation

10.8.3 EXPLORATION (20-25 MIN.)

Create tables and graphs from equations, identify which are linear, and write a conjecture about linear equations

10.8.2 COLLABORATION (20-25 MIN.)

Create tables of values from equations and match them to their graphs

10.8.4 WRAP-UP (~5 MIN.)

Select which equations are linear based on a conjecture

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Conjecture

MATERIALS

- Ruler
- Scientific calculator
- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- Consider allowing students to use a graphing utility.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make errors when using the equation to calculate the y-values for the x-values in a table.
- Students may confuse the x- and y-values when graphing the equation.
- Students may struggle to generalize that the exponent of the variable in linear equations is always one.

THINGS TO CONSIDER

- Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in creating a table of values from the equation of a function and graphing a function using a table of values.
- Consider allowing the students to use calculators. The lesson is an introduction to functions other than linear, so computational skills are not the focus.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

After graphing several equations in 10.8.3 Exploration, students are asked to notice and wonder about what the equations that are linear have in common and what the equations that are nonlinear have in common.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

The tables of values, graphs, and equations provide students with multiple representations of the function. During the lesson, students make connections between the representations and draw conclusions about how to determine if an equation is linear or nonlinear.

DIFFERENTIATE INSTRUCTION

For another entry point in 10.8.2 Collaboration, provide students with the table in question 2 and 4 first but include the x-values. After the students square or cube the x-values to the corresponding y-values, guide students back to questions 1 and 3 to find all values of x that are solutions to the equation.

Additionally, consider a Gallery Walk for 10.8.3 Exploration. Students can engage with each graph on chart paper and evaluate the work of others in a real-time graphing exercise that stimulates both thinking and motion.

10.8.1 WARM-UP: BIGGER AND SMALLER

DIFFERENTIATE INSTRUCTION

If students are struggling to get started or determine what the question is asking, consider prompting students to use one of their vocabulary words in their responses. Terms like “slope” or “y-intercept” in relation to largest and smallest might activate connections for students.

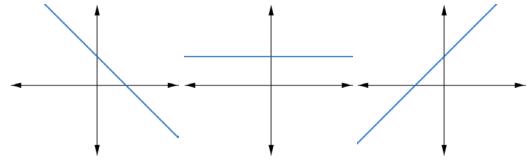
ENRICHMENT

How can the sign of the slope and the sign of the y-intercept be determined based on the position of the line?



10.8.1 Warm-Up: Bigger and Smaller

Three graphs are shown.



Diego said, “The graphs are ordered from smallest to largest.” Maya disagreed and said, “No. They are ordered from largest to smallest.”

But these are graphs, not numbers!

1. What do you think Diego and Maya are thinking?

Answers will vary.

Possible responses include:

- *Diego could have thought they were arranged from slope with the least value to slope with the greatest value.*
- *Maya could have been looking at where each graph started on the left side (this is called end behavior) from greatest to least.*



10.8.2 Collaboration: Revisiting Squares and Cubes

Work with your partner on the following.

1. Solve each equation for x . Show your work.

$$x^2 = 16$$

$$x = \pm 4$$

$$x^2 = 1$$

$$x = \pm 1$$

$$x^2 = 36$$

$$x = \pm 6$$

$$x^2 = 100$$

$$x = \pm 10$$

$$x^2 = 0$$

$$x = 0$$

2. Use your solutions to the previous equations to complete a table of values for the equation $y = x^2$.

x	-10	-6	-4	-1	0	1	4	6	10
y	100	36	16	1	0	1	16	36	100

3. Solve each equation for x . Show your work.

$$x^3 = 8$$

$$x = 2$$

$$x^3 = -64$$

$$x = -4$$

$$x^3 = -1$$

$$x = -1$$

$$x^3 = 125$$

$$x = 5$$

$$x^3 = 0$$

$$x = 0$$

$$x^3 = 64$$

$$x = 4$$

$$x^3 = -125$$

$$x = -5$$

$$x^3 = -8$$

$$x = -2$$

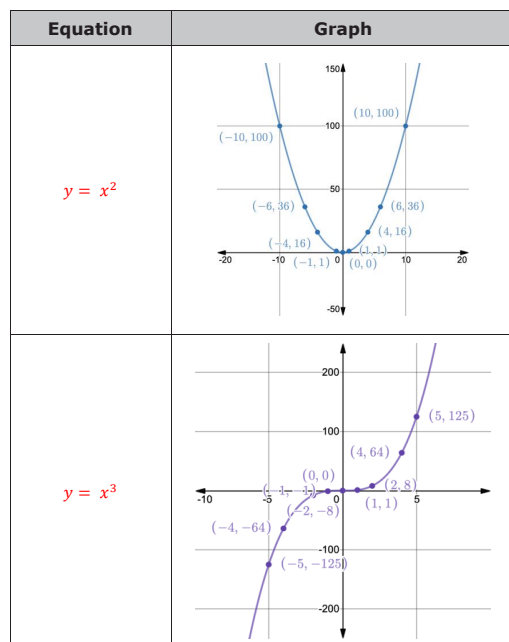
$$x^3 = 1$$

$$x = 1$$

4. Use your solutions to the previous equations to complete a table of values for the equation $y = x^3$.

x	-5	-4	-2	-1	0	1	2	4	5
y	-125	-64	-8	-1	0	1	8	64	125

5. The graphs of both equations are shown. Write the equation that matches each graph based on the tables of values created in the previous problems.



10.8.2 COLLABORATION: REVISITING SQUARES AND CUBES

DIFFERENTIATE INSTRUCTION

For another entry point, provide students with the x -values for the table in question 2. After the students square the x -values to find the corresponding y -values, guide students back to question 1 to find all values of x that solve the equation.

ENRICHMENT

- Why are all the y -values positive values?
- Why is there only one answer for $x^2 = 0$?

ENRICHMENT

- Why do the y -values include positive and negative values?
- Why is there only one answer for each of these equations?

QUESTIONING

Challenge advanced learners to explain why the graphs of these equations are nonlinear. Ask students to justify their conjectures by graphing additional equations with the features that they predict cause the equation to be nonlinear.

10.8.3 EXPLORATION: EQUATIONS AND THEIR GRAPHS

TEACHER MOVES

Conduct a Notice and Wonder routine with students to help them formulate generalizations about the common features of linear versus nonlinear equations. From these generalizations, students will write a conjecture about equations that are not linear.

DIFFERENTIATE INSTRUCTION

Consider a variety of methods and choose one that works best for your students and classroom.

- **Partner Work:** Students work in pairs, working collaboratively or with strategically assigned roles based on observational data as remediation. Students then work on questions 2-4 before finding two new partners for question 5.
- **Gallery Walk:** Sketch blank coordinate planes on chart paper and have students replicate this activity in groups on their own poster. Students can then Walk and Learn for questions 2-5.
- **Card Sort:** Prepopulate the tables and graphs and cut and copy into a set of cards. Each pair of students must match the correct tables, equations, and their corresponding graphs.
- **Alternative Sort:** Students each individually get their own card. They are not allowed to show anyone their card. Instead, students walk around the room, asking questions to find their pairing.

ENRICHMENT

- What do you notice about the sign of each of the y-values?
- How is this related to what quadrants the graph lies in?

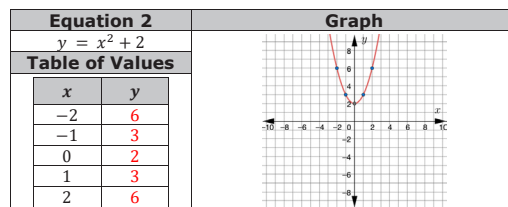
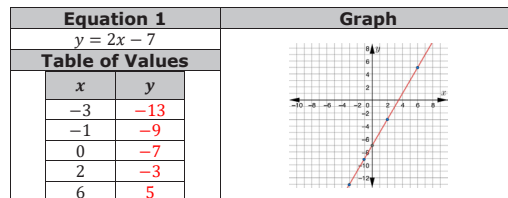


10.8.3 Exploration: Equations and Their Graphs

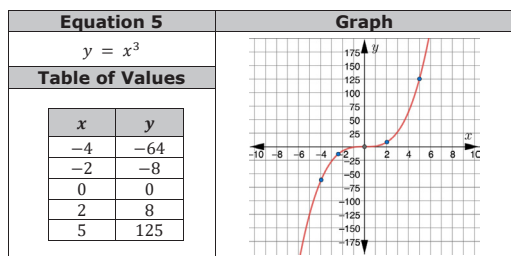
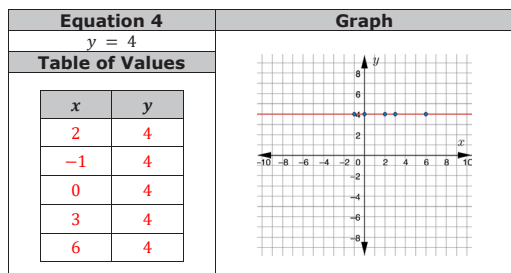
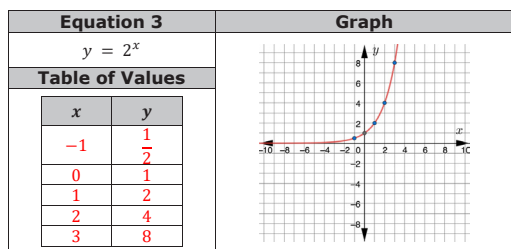
Relations between two variables can be represented using equations, tables of values, and graphs.

1. For each of the equations given, work with your partner to create a table of values. Then, plot each ordered pair from the table of values, using them to create a rough sketch of the graph by connecting the points.

One of the graphs and its corresponding table from the collaboration activity have already been included.



10.8.3 EXPLORATION: EQUATIONS AND THEIR GRAPHS (CONTINUED)



TEACHER MOVES

If students don't see the shape of the graph from the x -values they chose, have the students choose more or different x -values. Students may need help with negative x -values but shouldn't avoid them, since this will determine the shape of the graph. Students may need to look back at Unit 4 for a refresher on negative exponents.

ENRICHMENT

What is the difference between $2x$ and 2^x ?

ENRICHMENT

What do you notice about the y -values? Why do you think this is true?

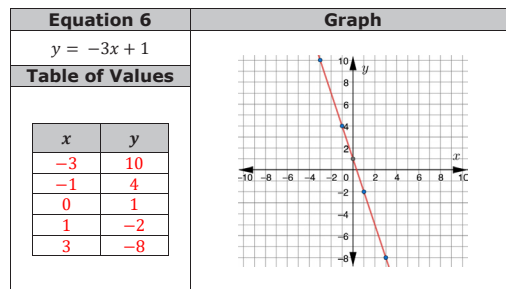
TEACHER MOVES

Students completed this table in question 4 in 10.8.2 Collaboration. The graph was also shown in question 5 in 10.8.2 Collaboration. It is placed here in the same format as the other equations to help students with their conjecture.

10.8.3 EXPLORATION: EQUATIONS AND THEIR GRAPHS (CONTINUED)

TEACHER MOVES

Students may identify $y = 4$ as non-linear, reasoning that $y = 4$ is a point on the plane. If so, prompt them to identify a location on the coordinate plane where $y = 4$. Most students will likely choose $(0,4)$. If so, mark other points like $(1,4)$ or $(-2,4)$ and ask questions like “Does $y=4$ here as well? How do you know? If $y=4$ here as well, and here, connect those points – what do you notice?”



2. Use the table below to record your answers to the following based on the exploration.

Equation 1 * $y = 2x - 7$	Equation 2 $y = x^2 + 2$	Equation 3 $y = 2^x$
Equation 4 * $y = 4$	Equation 5 $y = x^3$	Equation 6 * $y = -3x + 1$

- Put an asterisk (*) next to each equation that corresponds to a linear graph.
- Circle each equation that corresponds to a graph that is NOT linear.

ENRICHMENT

Notice and Wonder

Consider framing question 3 as a Notice and Wonder. Students should write their answers in each box, making note of what they have in common to then make a conjecture in question 4. Encourage students that any Notice or anything they have in common can be helpful and that there is no one correct answer here.

DIFFERENTIATE INSTRUCTION

Consider a foldable, Frayer Model, or type of interactive notebook for this summary.

Examples and verbal descriptions of linear and nonlinear equations will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

3. Compare the equations that are linear to those that are not linear.

What do the equations that are linear have in common?	What do the equations that are NOT linear have in common?
Answers will vary. Possible responses include noticing that the exponents of the variables are 1	Answers will vary. Possible responses include noticing that the equation either has a variable with an exponent other than 1 or the variable is the exponent.

4. Make a conjecture about the types of equations that are not linear.
Answers will vary. A possible conjecture may say “Equations with variable exponents other than 1 and/or equations with variables in the exponent are not linear.”

5. Ask two classmates their conjecture. Record their conjectures and write your summary of their reasons. You are the only person who should write in the first two columns of the table. Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Conjecture	My Summary of Their Reason	Initials



10.8.4 Wrap-Up: Cool Down

1. Eleven equations are shown. Based on your conjecture about the types of equations that are not linear, circle the equations you *predict* will produce a graph that is **not** a line. Use your best judgement. You will have an opportunity to make revisions during the next lesson.

$$y = -2$$

$$y = \frac{1}{2}\sqrt{x} - 1$$

$$\frac{x^3}{2} = y$$

$$y = -x^2 + 6$$

$$5x = 6 + 3y$$

$$x = 5$$

$$y = \frac{3}{5}x - 7$$

$$x = y^2$$

$$y = 3^x - 2$$

$$y = \frac{2}{x} + 1$$

$$y = |x| - 5$$

10.8.4 WRAP-UP: COOL DOWN

TEACHER MOVES

This activity is a prediction to build on their conjectures from the 10.8.3 Exploration. Students are not expected at this point to have an understanding of all of the functions. Teachers can collect from the students the functions that they circled as not linear for discussion during Lesson 9.